VUA-SEN307 Lab 2: Normalization (1NF to 3NF)

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0. Unnormalized Table

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems, Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng., Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems, Operating Systems	Dr A. Okafor, Mrs F. Eze	Software Eng., Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

1. First Normal Form (1NF) **Primary key: (matric_no, course_code)**

matric_no	student_name	student_dept	course_code	course_title	lecturer	lecturer_dept	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

Fixed: I split the lists so every cell holds one value, giving each student-course pair its own row.

2. Second Normal Form (2NF)

Student (PK: matric_no)

matric_no	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Course (PK: course_code)

course_code	course_title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

Enrolment (PK: matric_no + course_code)

matric_no	course_code	session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

Course_Lecturer (PK: course_code + lecturer)

course_code	lecturer	lecturer_dept	session
SEN307	Dr A. Okafor	Software Eng.	2026/2027
SEN301	Dr M. Bello	Software Eng.	2026/2027
COS301	Mrs F. Eze	Computer Sci.	2026/2027

Fixed: I removed partial dependencies, so student details depend only on matric_no, and course and lecturer details depend only on the course, not on the full (matric_no, course_code) key.

3. Third Normal Form (3NF)

Department (PK: dept_id)

dept_id	dept_name
1	Software Engineering
2	Computer Science

Student (PK: student_id; FK: dept_id)

student_id	matric_no	full_name	dept_id
1	20231234	Ada Obi	1
2	20231235	John Musa	2
3	20231236	Grace Okon	1

Lecturer (PK: lecturer_id; FK: dept_id)

lecturer_id	lecturer_name	dept_id
1	Dr A. Okafor	1
2	Dr M. Bello	1
3	Mrs F. Eze	2

Course (PK: course_id)

course_id	course_code	course_title
1	SEN307	Database Systems
2	SEN301	Software Architecture
3	COS301	Operating Systems

Enrolment (PK: enrolment_id; FK: student_id, course_id)

enrolment_id	student_id	course_id	session
1	1	1	2026/2027
2	1	2	2026/2027
3	2	1	2026/2027
4	2	3	2026/2027
5	3	2	2026/2027

Teaching (PK: teaching_id; FK: course_id, lecturer_id)

teaching_id	course_id	lecturer_id	session
1	1	1	2026/2027
2	2	2	2026/2027
3	3	3	2026/2027

Fixed: I removed transitive dependencies by moving department names into their own Department table (also fixing "Software Engineering" vs "Software Eng."), so renaming a department is now one update.

4. Final Schema

```
Department (dept_id PK, dept_name)
Student    (student_id PK, matric_no, full_name, dept_id FK -> Department)
Lecturer   (lecturer_id PK, lecturer_name, dept_id FK -> Department)
Course     (course_id PK, course_code, course_title)
Enrolment  (enrolment_id PK, student_id FK -> Student, course_id FK -> Course, session)
Teaching   (teaching_id PK, course_id FK -> Course, lecturer_id FK -> Lecturer, session)
```

*Student and Course are many-to-many, resolved by **Enrolment**. Course and Lecturer are many-to-many, resolved by **Teaching**.*